ITSIMPLERA INSTITUTE

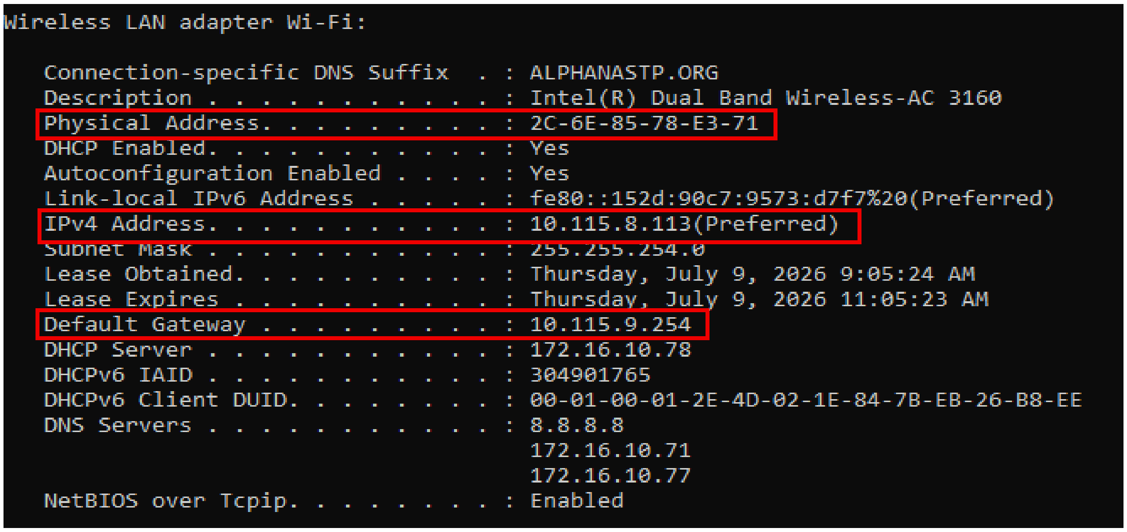
Cybersecurity Analyst Internship

RED TEAM ASSIGNMENT # 02

PREPARED BY	AMNA IMTIAZ
REGISTRATION NO.	REDBO1 - 2350
SUPERVISOR	TALAL REHAN
DATE OF SUBMISSION	9th July 2026

QUESTION # 01

Network Information



```
Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . : ALPHANASTP.ORG
   Description . . . . . . . . . . . : Intel(R) Dual Band Wireless-AC 3160
   Physical Address. . . . . . . . . : 2C-6E-85-78-E3-71
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::152d:90c7:9573:d7f7%20(Preferred)
   IPv4 Address. . . . . . . . . . . : 10.115.8.113(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.254.0
   Lease Obtained. . . . . . . . . . : Thursday, July 9, 2026 9:05:24 AM
   Lease Expires . . . . . . . . . . : Thursday, July 9, 2026 11:05:23 AM
   Default Gateway . . . . . . . . . : 10.115.9.254
   DHCP Server . . . . . . . . . . . : 172.16.10.78
   DHCPv6 IAID . . . . . . . . . . . : 304901765
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-2E-4D-02-1E-84-7B-EB-26-B8-EE
   DNS Servers . . . . . . . . . . . : 8.8.8.8
                                       172.16.10.71
                                       172.16.10.77
   NetBIOS over Tcpip. . . . . . . . : Enabled
```

1. IPv4 Address

Value: 10.115.8.113

Purpose

The IPv4 address is the unique logical address assigned to your computer on the network. It enables your device to communicate with other devices and access the internet.

2. MAC Address (Physical Address)

Value: 2C-6E-85-78-E3-71

Purpose

The MAC (Media Access Control) address is the unique hardware address of your Wi-Fi network adapter. It is used to identify your device on the local network.

3. Default Gateway

Value: 10.115.9.254

Purpose

The default gateway is the IP address of the router that forwards data from your local network to other networks, including the internet.

QUESTION # 02

Basic Networking Commands

1. ipconfig /all

Command: ipconfig /all

```
C:\Users\UsEr>ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : DESKTOP-HKHTI5F
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Mixed
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No
   DNS Suffix Search List. . . . . . : ALPHANASTP.ORG

Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Realtek PCIe FE Family Controller
   Physical Address. . . . . . . . . : 84-7B-EB-26-B8-EE
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Ethernet adapter Ethernet 2:
```

Purpose

Displays detailed network configuration, including IP address, MAC address, DNS server, subnet mask, and default gateway.

2. ping google.com

Command: ping google.com

```
C:\Users\UsEr>ping google.com

Pinging google.com [142.250.203.174] with 32 bytes of data:
Reply from 142.250.203.174: bytes=32 time=40ms TTL=114
Reply from 142.250.203.174: bytes=32 time=39ms TTL=114
Reply from 142.250.203.174: bytes=32 time=38ms TTL=114
Request timed out.

Ping statistics for 142.250.203.174:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 38ms, Maximum = 40ms, Average = 39ms
```

Purpose

Tests connectivity between your computer and Google's server. It also measures response time and checks whether packets are successfully transmitted.

3. tracert google.com

Command: tracert google.com

```
C:\Users\UsEr>tracert google.com

Tracing route to google.com [142.250.203.174]
over a maximum of 30 hops:

  1   104 ms    26 ms    75 ms  10.115.9.254
  2     2 ms     3 ms     2 ms  10.115.241.10
  3     4 ms     *        3 ms  115-186-173-125.nayatel.pk [115.186.173.125]
  4     *        5 ms     6 ms  172.31.8.41
  5     *        4 ms     *     172.31.8.29
  6     6 ms     6 ms     4 ms  172.31.5.170
  7    23 ms     *       23 ms  110.93.202.134
  8    23 ms    24 ms     *     110.93.253.110
  9     *       25 ms    23 ms  110.93.252.246
 10     *       40 ms    39 ms  110.93.192.207
 11    43 ms    42 ms    42 ms  172.253.51.55
 12    40 ms     *        *     142.251.50.215
 13     *       41 ms    45 ms  pnfjra-ah-in-f14.1e100.net [142.250.203.174]

Trace complete.
```

Purpose

Shows the path (hops) that packets travel from your computer to Google's server. It helps identify where network delays or failures occur.

4. nslookup google.com

Command: nslookup google.com

```
C:\Users\UsEr>nslookup google.com
Server:  dns.google
Address:  8.8.8.8

DNS request timed out.
    timeout was 2 seconds.
DNS request timed out.
    timeout was 2 seconds.
Non-authoritative answer:
Name:    google.com
Addresses:  2a00:1450:4019:817::200e
          142.250.203.174
```

Purpose

Queries the Domain Name System (DNS) to translate the domain name into its corresponding IP address.

5. arp -a

Command: arp -a

```
C:\Users\UsEr>arp -a

Interface: 192.168.56.1 --- 0xa
  Internet Address      Physical Address      Type
  192.168.56.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.2             01-00-5e-00-00-02     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static

Interface: 10.115.8.113 --- 0x14
  Internet Address      Physical Address      Type
  10.115.8.65           18-56-80-ec-24-ad     dynamic
  10.115.8.115          e4-a4-71-17-1e-cf     dynamic
  10.115.9.252          c8-b6-d3-59-ee-83     dynamic
  10.115.9.254          dc-ef-80-ba-0f-69     dynamic
  10.115.9.255          ff-ff-ff-ff-ff-ff     static
  224.0.0.2             01-00-5e-00-00-02     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static
```

Purpose

Displays the ARP (Address Resolution Protocol) table, showing the mapping between IP addresses and MAC addresses of devices on the local network.

QUESTION # 03

Router vs. Switch

Feature	Router	Switch
Primary Function	Connects different networks	Connects devices within the same network
OSI Layer	Layer 3 (Network Layer)	Layer 2 (Data Link Layer)
Address Used	IP Address	MAC Address
Data Forwarding	Routes packets between networks	Forwards frames within a LAN
Internet Access	Provides internet connectivity	Does not directly provide internet access
Common Use Case	Home/office router connecting LAN to Internet	Connecting computers, printers, and servers in a LAN
Figure		

QUESTION # 04

IPv4 vs. IPv6

Feature	IPv4	IPv6
Address Length	32 bits	128 bits
Address Format	Decimal	Hexadecimal
Number of Addresses	About 4.3 billion	About 340 undecillion
Header Size	Variable (20-60 bytes)	Fixed (40 bytes)
Security	IPsec optional	IPsec support built-in
Configuration	Manual or DHCP	SLAAC or DHCPv6

Example IPv4 Address

192.168.1.10

Example IPv6 Address

2001:0db8:85a3:0000:0000:8a2e:0370:7334

```
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   IPv4 Address. . . . . . . . . . . : 10.115.8.113
   Subnet Mask . . . . . . . . . . . : 255.255.254.0
   Default Gateway . . . . . . . . . : 10.115.9.254
```

QUESTION # 05

OSI Model

The OSI (Open Systems Interconnection) Model consists of seven layers that define how data is transmitted between devices over a network. The layers, from top to bottom, are:

7. Application Layer

Primary Function: Provides network services directly to end-user applications.
Common Protocol: HTTP
Networking Device: Proxy Server

6. Presentation Layer

Primary Function: Translates, encrypts/decrypts, and compresses data before transmission.
Common Protocol: SSL/TLS
Networking Device: Gateway

5. Session Layer

Primary Function: Establishes, manages, and terminates communication sessions between devices.
Common Protocol: NetBIOS
Networking Device: Gateway

4. Transport Layer

Primary Function: Ensures reliable end-to-end data delivery, flow control, and error recovery.
Common Protocol: TCP
Networking Device: Firewall

3. Network Layer

Primary Function: Determines the best path for data and routes packets between networks.
Common Protocol: IP (Internet Protocol)
Networking Device: Router

2. Data Link Layer

Primary Function: Transfers data frames using MAC addresses and detects transmission errors.
Common Protocol: Ethernet (IEEE 802.3)
Networking Device: Switch

1. Physical Layer

Primary Function: Transmits raw bits over the physical transmission medium, such as cables or wireless signals.
Common Protocol: Ethernet Physical Standard
Networking Device: Hub